

The SVM approach

- We have briefly discussed Support Vector Machine (SVM) idea.
- The idea is to map the feature vectors nonlinearly into another space and learn a linear classifier there.
- The linear classifier in this new space would be an appropriate nonlinear classifier in the original space.



- Recall the simple example we saw earlier.
- Let $X = [x_1 \ x_2]$ and let $\phi : \mathbb{R}^2 \rightarrow \mathbb{R}^5$ given by

$$Z = \phi(X) = [1 \ x_1 \ x_2 \ x_1^2 \ x_2^2 \ x_1x_2]$$

- Now,

$$g(X) = a_0 + a_1x_1 + a_2x_2 + a_3x_1^2 + a_4x_2^2 + a_5x_1x_2$$

is a quadratic discriminant function in $\mathbb{R}^2$; but

$$g(Z) = a_0 + a_1z_1 + a_2z_2 + a_3z_3 + a_4z_4 + a_5z_5$$

is a linear discriminant function in the ' $\phi(X)$ ' space.



- There are two major issues in naively using this idea.
- If we want, e.g., p^{th} degree polynomial discriminant function in the original feature space ($\mathbb{R}^m$), then the transformed feature vector, Z , has dimension $O(m^p)$.
- Results in huge computational cost both for learning and and final operation of the classifier.
- We need to learn $O(m^p)$ parameters rather than $O(m)$ parameters. Hence may need much larger number of examples for achieving proper generalization.



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- SVM offers an elegant solution to both.



Support Vector Machines

- Learning of optimal hyperplane.
 - Separating hyperplane that maximizes separation between Classes.
- *Effectively* maps original feature vectors into a high dimensional space. Hence learns nonlinear discriminant functions.
- By using Kernel function we never need to explicitly calculate the mapping.
- We need solve only a quadratic optimization problem.
- Now we formulate the SVM method, first for linearly separable case.



- Training set:

$$\{(X_i, y_i), \quad i = 1, \dots, n\}, \quad X_i \in \mathbb{R}^m, \quad y_i \in \{+1, -1\}.$$

- To start with, assume training set is linearly separable.
That is, exist $W \in \mathbb{R}^m$ and $b \in \mathbb{R}$ such that

$$W^T X_i + b > 0, \quad \forall i \text{ s.t. } y_i = +1$$

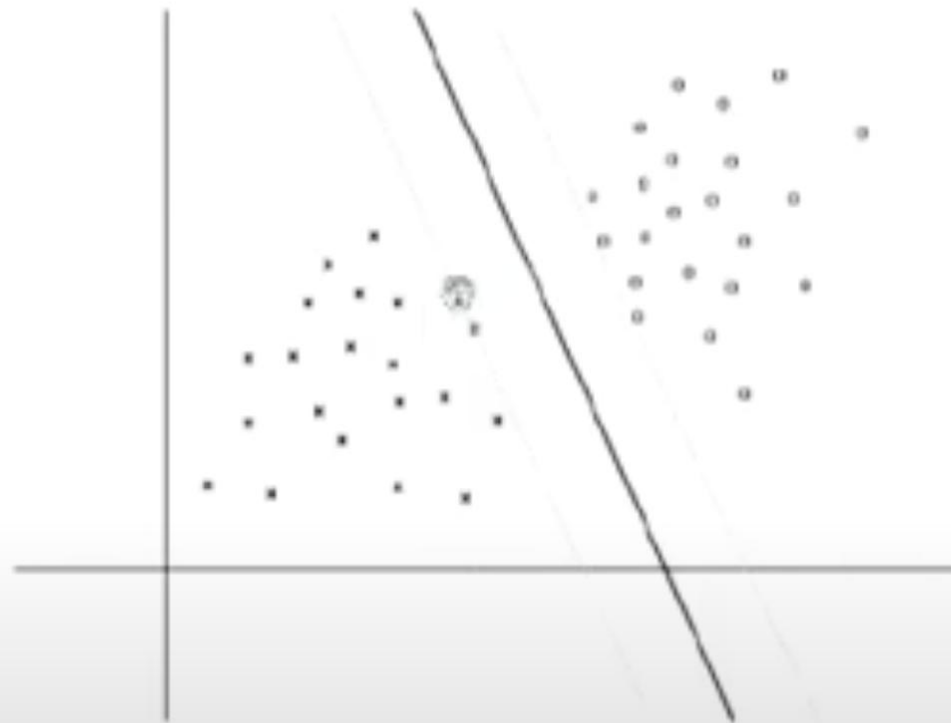
$$W^T X_i + b < 0, \quad \forall i \text{ s.t. } y_i = -1$$

(Note both inequalities are strict)

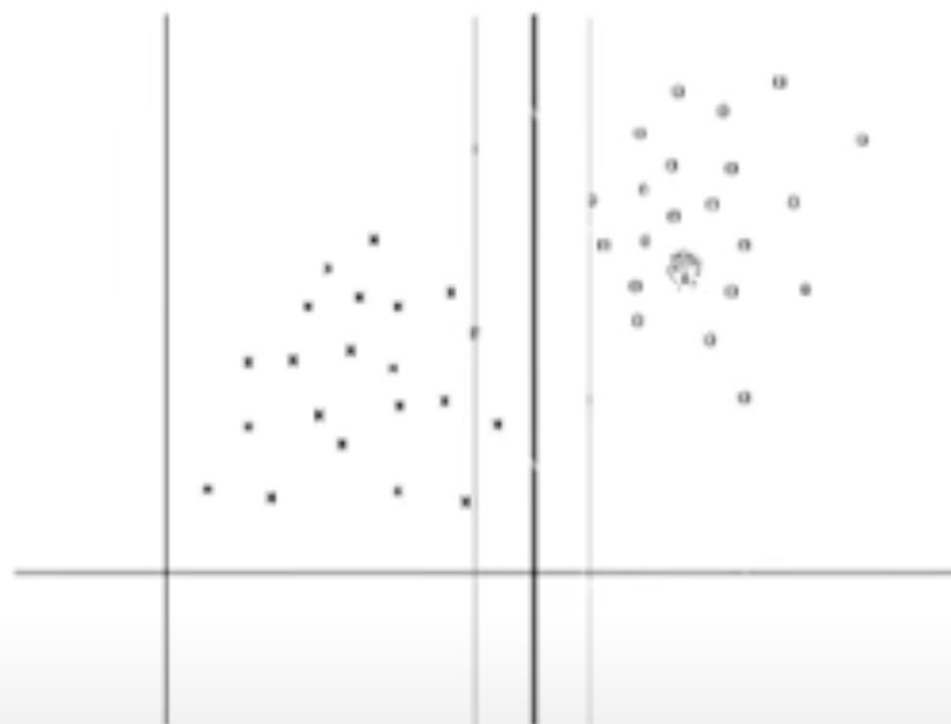
- $W^T X + b = 0$ – A separating hyperplane.
- Infinitely many separating hyperplanes exist.



A good separating hyperplane



Another separating hyperplane



- Recall that we assume training set is linearly separable and hence

$$W^T X_i + b > 0, \quad \forall i \text{ s.t. } y_i = +1$$

$$W^T X_i + b < 0, \quad \forall i \text{ s.t. } y_i = -1$$

- Since the training set is finite, $\exists \epsilon > 0$ s.t.

$$W^T X_i + b \geq \epsilon, \quad \forall i \text{ s.t. } y_i = +1$$

$$W^T X_i + b \leq -\epsilon, \quad \forall i \text{ s.t. } y_i = -1$$



- Hence, we can scale W , b such that

$$W^T X_i + b \geq +1 \text{ if } y_i = +1$$

$$W^T X_i + b \leq -1 \text{ if } y_i = -1$$

or, equivalently

$$y_i(W^T X_i + b) \geq 1, \quad \forall i.$$

(Recall that $y_i \in \{+1, -1\}$)



- When the training set is separable, any separating hyperplane, W , b , can be scaled to satisfy

$$y_i(W^T X_i + b) \geq 1, \quad \forall i.$$

- Then there are no training patterns between the two parallel hyperplanes

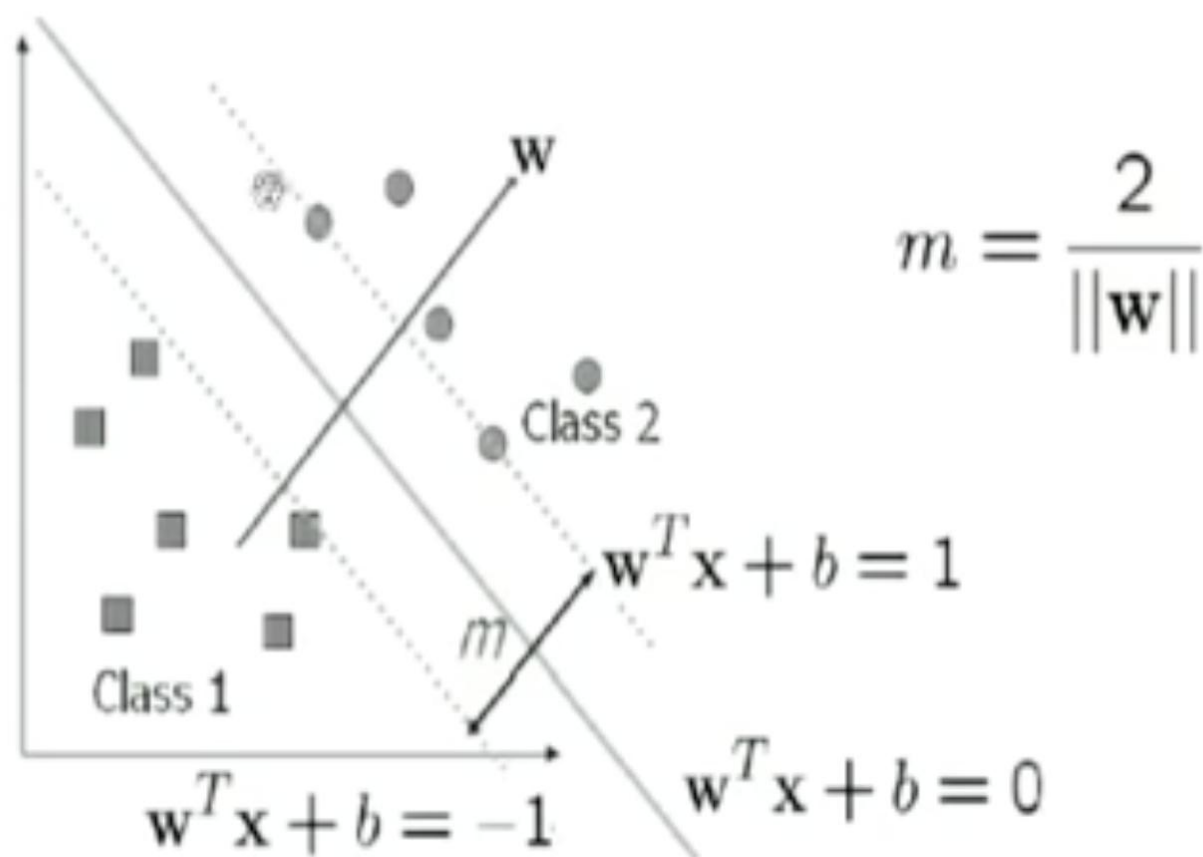
$$W^T X + b = +1$$

and

$$W^T X + b = -1$$



Margin of a hyperplane



Optimal hyperplane

- Distance between these two hyperplanes is: $\frac{2}{||W||}$.
Called margin of the separating hyperplane.
- Hence distance between the hyperplane and the closest pattern is $\frac{1}{||W||}$.
- Intuitively, more the margin, better is the chance of correct classification of new patterns.
- **Optimal Hyperplane** – separating hyperplane with maximum margin.



The optimization problem

- Among all separating hyperplanes, the one with largest margin is the optimal hyperplane.
- So, the optimal hyperplane is a solution to the following optimization problem.
- Find $W \in \mathbb{R}^m$, $b \in \mathbb{R}$ to

$$\begin{array}{ll}\text{minimize} & \frac{1}{2} W^T W \\ \text{subject to} & y_i(W^T X_i + b) \geq 1, \quad i = 1, \dots, n\end{array}$$

- This is a constrained optimization problem with quadratic cost function and linear inequality constraints.



Constrained Optimization

- Consider the following optimization problem

$$\begin{array}{ll}\text{minimize} & f(\mathbf{x}) \\ \text{subject to} & \mathbf{a}_j^T \mathbf{x} + b_j \leq 0, \quad j = 1, \dots, r\end{array}$$

where $f : \mathbb{R}^m \rightarrow \mathbb{R}$ is a continuously differentiable function, and
 $\mathbf{a}_j \in \mathbb{R}^m, b_j \in \mathbb{R}, j = 1, \dots, r$.

- A point, $\mathbf{x} \in \mathbb{R}^m$, is called a **feasible point** (for this problem) if $\mathbf{a}_j^T \mathbf{x} + b_j \leq 0, j = 1, \dots, r$.



- Any $\mathbf{x}^* \in \mathcal{R}^m$ is called a local minimum of the problem if $f(\mathbf{x}^*) \leq f(\mathbf{x})$ for all $\mathbf{x}$ that is feasible and is in a small neighbourhood of $\mathbf{x}^*$.
- If $f(\mathbf{x}^*) \leq f(\mathbf{x}), \forall \mathbf{x} \in \mathcal{R}^m$ and $\mathbf{x}$ feasible, then $\mathbf{x}^*$ is a global minimum.
- Unlike in unconstrained optimization, here we need to minimize only over the feasible set.



- Here we would consider only the case where f is a convex function.
- $f : \mathbb{R}^m \rightarrow \mathbb{R}$ is said to be a convex function if for all $\mathbf{x}_1, \mathbf{x}_2 \in \mathbb{R}^m$ and for all $\alpha \in (0, 1)$,

$$f(\alpha \mathbf{x}_1 + (1 - \alpha) \mathbf{x}_2) \leq \alpha f(\mathbf{x}_1) + (1 - \alpha) f(\mathbf{x}_2)$$

- For example, $f(\mathbf{x}) = \mathbf{x}^T \mathbf{x}$ is a convex function.
- When f is convex, in our optimization problem, every local minimum is also a global minimum.



- We now look at one method of solving the constrained optimization problem.
- Given our optimization problem, define

$$L(\mathbf{x}, \boldsymbol{\mu}) = f(\mathbf{x}) + \sum_{j=1}^r \mu_j (\mathbf{a}_j^T \mathbf{x} + b_j)$$

- The L is called the Lagrangian of the problem and the μ_j are called the Lagrange multipliers.
- Essentially, the constrained optimization problem can be solved through unconstrained optimization of L .



Kuhn-Tucker Conditions

- Consider the optimization problem with f convex.
- Any $\mathbf{x}^*$ is a global minimum if and only if $\mathbf{x}^*$ is feasible and there exist μ_j^* , $j = 1, \dots, r$, such that
 1. $\nabla_x L(\mathbf{x}^*, \boldsymbol{\mu}^*) = 0$
 2. $\mu_j^* \geq 0, \quad \forall j$
 3. $\mu_j^*(\mathbf{a}_j^T \mathbf{x}^* + b_j) = 0, \quad \forall j$
- These are the so called Kuhn-Tucker conditions for our optimization problem with convex cost function and linear constraints.



- We can use the above conditions to obtain a x^* which is a minimum of the optimization problem.
- We can also solve the constrained optimization problem using the so called dual of this problem.
- This is the approach taken in SVM algorithm.
- Duality is an important concept in optimization.
- Here we discuss only one way of formulating the dual which is useful when the objective function is convex and constraints are linear.



- Our optimization problem is

$$\begin{array}{ll}\text{minimize} & f(\mathbf{x}) \\ \text{subject to} & \mathbf{a}_j^T \mathbf{x} + b_j \leq 0, \quad j = 1, \dots, r\end{array}$$

where $f : \mathbb{R}^m \rightarrow \mathbb{R}$ is a continuously differentiable convex function, and

$$\mathbf{a}_j \in \mathbb{R}^m, b_j \in \mathbb{R}, j = 1, \dots, r.$$

- This is known as the **primal** problem.
- Here the optimization variables are $\mathbf{x} \in \mathbb{R}^m$.



- Recall that the Lagrangian is

$$L(\mathbf{x}, \boldsymbol{\mu}) = f(\mathbf{x}) + \sum_{j=1}^r \mu_j (\mathbf{a}_j^T \mathbf{x} + b_j)$$

Here, $\mathbf{x} \in \mathbb{R}^m$ and $\boldsymbol{\mu} \in \mathbb{R}^r$.

- Define the *dual function*, $q : \mathbb{R}^r \rightarrow [-\infty, \infty)$ by

$$q(\boldsymbol{\mu}) = \inf_{\mathbf{x}} L(\mathbf{x}, \boldsymbol{\mu}) \quad (2)$$

- If for a particular $\boldsymbol{\mu}$, if the infimum is not attained then $q(\boldsymbol{\mu})$ would take value $-\infty$.



The Dual problem

- The dual problem is:

$$\begin{array}{ll}\text{maximize} & q(\boldsymbol{\mu}) \\ \text{subject to} & \mu_j \geq 0, \quad j = 1, \dots, r\end{array}$$

- This is also a constrained optimization problem.
- Here the optimization is over $\Re^r$ and $\boldsymbol{\mu} \in \Re^r$ are the optimization variables.
- There is a nice connection between the primal and dual problems.



Primal-Dual Relationship

- Now we have the following.
 1. If the primal has a solution so does the dual and the optimal values are equal.
 2. $\mathbf{x}^*$ is optimal for primal and $\boldsymbol{\mu}^*$ is optimal for dual if and only if
 - $\mathbf{x}^*$ is feasible for primal and $\boldsymbol{\mu}^*$ is feasible for dual,
 - $f(\mathbf{x}^*) = L(\mathbf{x}^*, \boldsymbol{\mu}^*) = \min_{\mathbf{x}} L(\mathbf{x}, \boldsymbol{\mu}^*)$.
- We would be using the dual formulation for the optimization problem in SVM



The optimization problem for SVM

- The optimal hyperplane is a solution of the following constrained optimization problem.
- Find $W \in \mathbb{R}^m$, $b \in \mathbb{R}$ to

$$\text{minimize} \quad \frac{1}{2}W^T W$$

$$\text{subject to} \quad 1 - y_i(W^T X_i + b) \leq 0, \quad i = 1, \dots, n$$

- Quadratic cost function and linear (inequality) constraints.
- Kuhn-Tucker conditions are necessary and sufficient. Every local minimum is global minimum.



- The Lagrangian is given by

$$L(W, b, \boldsymbol{\mu}) = \frac{1}{2}W^T W + \sum_{i=1}^n \mu_i [1 - y_i(W^T X_i + b)]$$

- The Kuhn-Tucker conditions give

$$\nabla_W L = 0 \Rightarrow W^* = \sum_{i=1}^n \mu_i^* y_i X_i$$

$$\frac{\partial L}{\partial b} = 0 \Rightarrow \sum_{i=1}^n \mu_i^* y_i = 0$$

$$1 - y_i(X_i^T W^* + b^*) \leq 0, \quad \forall i$$

$$\mu_i^* \geq 0, \quad \& \quad \mu_i^* [1 - y_i(X_i^T W^* + b^*)] = 0, \quad \forall i$$



- Let $S = \{i \mid \mu_i^* > 0\}$.
- By complementary slackness condition,

$$i \in S \quad \Rightarrow \quad y_i(X_i^T W^* + b^*) = 1$$

Implies X_i is closest to separating hyperplane.

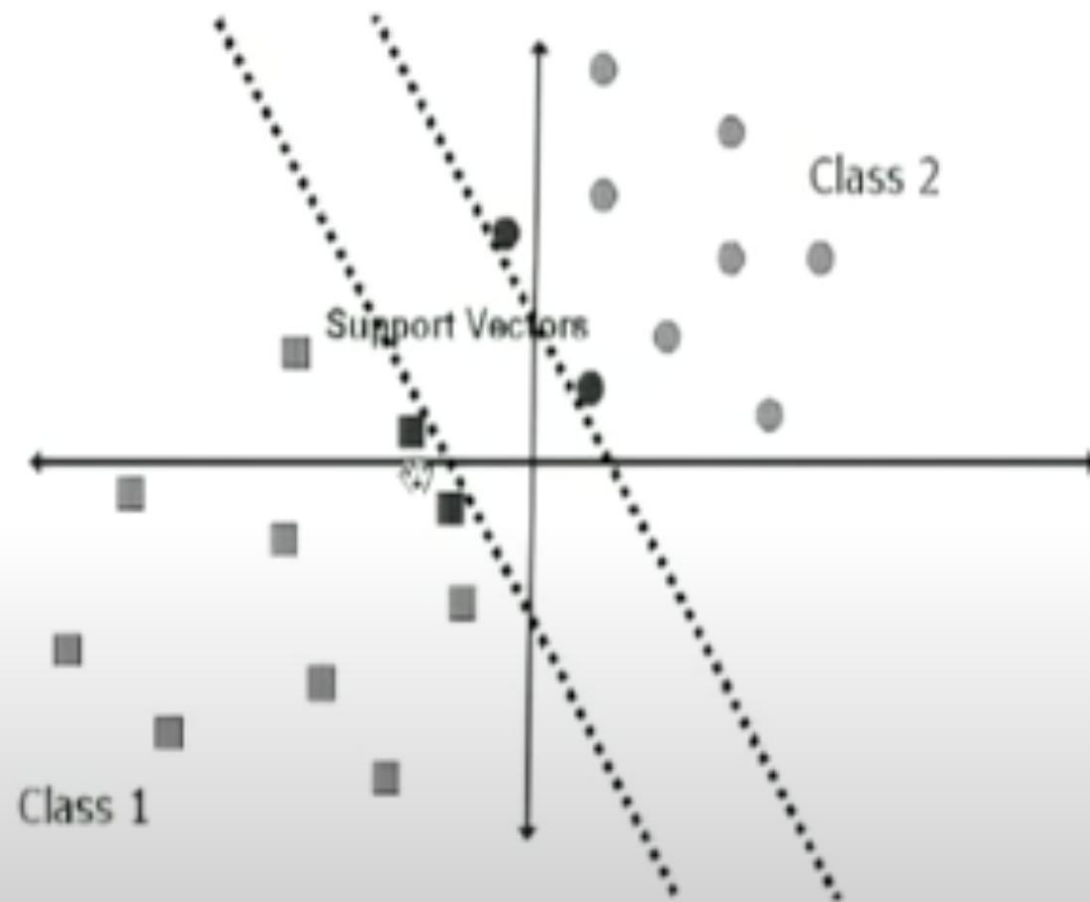
- $\{X_i \mid i \in S\}$ are called Support vectors. We have

$$W^* = \sum_i \mu_i^* y_i X_i = \sum_{i \in S} \mu_i^* y_i X_i \quad \text{⑩}$$

- Optimal W is a linear combination of Support vectors.
- Support vectors constitute a very useful output of the method.



Optimal hyperplane



The SVM solution

- The optimal hyperplane – W^* , b^* given by:

$$W^* = \sum_i \mu_i^* y_i X_i = \sum_{i \in S} \mu_i^* y_i X_i$$

$$b^* = y_j - X_j^T W^*, \quad j \text{ s.t. } \mu_j^* > 0$$

(Note that $\mu_j^* > 0 \Rightarrow y_j(X_j^T W^* + b^*) = 1$)

- Thus, W^* , b^* are determined by μ_i^* , $i = 1, \dots, n$.
- We can use the dual of the optimization problem to get μ_i^* .



Dual optimization problem for SVM

- The dual function is

$$q(\boldsymbol{\mu}) = \inf_{W,b} \left\{ \frac{1}{2} W^T W + \sum_{i=1}^n \mu_i [1 - y_i (W^T X_i + b)] \right\}$$

- If $\sum \mu_i y_i \neq 0$ then $q(\boldsymbol{\mu}) = -\infty$.
- Hence we need to maximize q only over those $\boldsymbol{\mu}$ s.t. $\sum \mu_i y_i = 0$.
- Infimum w.r.t. W is attained at $W = \sum \mu_i y_i X_i$.
- We obtain the dual by substituting $W = \sum \mu_i y_i X_i$ and imposing $\sum \mu_i y_i = 0$.



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- By substituting $W = \sum \mu_i y_i X_i$ and $\sum \mu_i y_i = 0$ we get

$$\begin{aligned} q(\boldsymbol{\mu}) &= \frac{1}{2} W^T W + \sum_{i=1}^n \mu_i - \sum_{i=1}^n \mu_i y_i (W^T X_i + b) \\ &= \frac{1}{2} \left(\sum_i \mu_i y_i X_i \right)^T \sum_j \mu_j y_j X_j + \sum_i \mu_i \\ &\quad - \sum_i \mu_i y_i X_i^T \left(\sum_j \mu_j y_j X_j \right) \\ &= \sum_i \mu_i - \frac{1}{2} \sum_i \sum_j \mu_i y_i \mu_j y_j X_i^T X_j \end{aligned}$$



- Thus, the dual problem is:

$$\max_{\boldsymbol{\mu}} \quad q(\boldsymbol{\mu}) = \sum_{i=1}^n \mu_i - \frac{1}{2} \sum_{i,j=1}^n \mu_i \mu_j y_i y_j X_i^T X_j$$

subject to $\mu_i \geq 0, \quad i = 1, \dots, n, \quad \sum_{i=1}^n y_i \mu_i = 0$

- Quadratic cost function and linear constraints
- Training data vectors appear only as innerproduct
- Optimization is over $\mathbb{R}^n$ irrespective of the dimension of X_i .

